Are you prepared to face monsoon induced landslides?



The southwest monsoon has arrived. With the torrential rains that can come with it, the landslide risk in the country is expected to increase. Typically, this monsoon brings rains to districts covering the wet-zone of the country. Accordingly, the parts of the country which are to experience landslide and cutting failure disasters during this monsoon would be Hambantota, Matara, Galle, Kegalle, Rathnapura, Kalutara, Colombo, Gampaha, Kururnegala districts, western slopes of Nuwaraeliya district including Ambagamuwa Korale and Kotmale DSDs, Ganga Ihala Korale, Pasbage DSDs and certain parts of Udapalatha DSD of Kandy district and certain parts of Laggala, Weligamuwa, Ukuwela DSDs of Matale district.

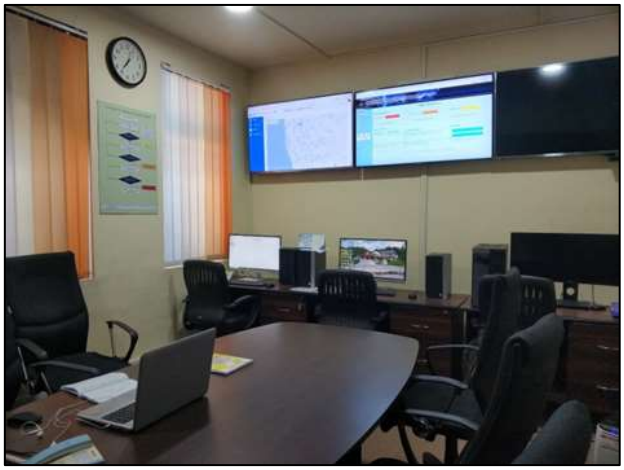


Landslides and cutting failures

Taking the situation into consideration, the National Building Research Organisation has taken measures to successfully face any upcoming landslide risk circumstances.

1. ***Landslide Early Warnings***:
   Landslide early warnings are issued in two categories; as regional and site-specific.

NBRO issues regional early warnings at DSD level using real time rainfall data received through a network of over 250 automated rain gauges established in landslide mountainous areas of the country. For this purpose a landslide early warning center has been established within NBRO office premises with facilities to analyse rainfall data. A workforce consisting of scientists from the Landslide Research and Risk Management Division of NBRO contributes to smooth functioning of this early warning center 24/7. In addition, the Landslide Research and Risk Management Division of NBRO is constantly carrying out research which will support further improvement of the early warning process in terms of accuracy and ease.



Landslide Early Warning Center of NBRO

Site specific landslide early warnings are issued for places with pre-identified landslide risk. Two mechanisms are in use for the process. One mechanism is to make use of signals give out by automated landslide detection equipment mounted on ground. The other mechanism is the community based approach in which the residents of the risk equipped with manual rain gauges measure the rainfall themselves and take decisions to take appropriate measures as the rainfall reaches danger levels or self-evacuate when needed.



Automated equipment used for site specific landslide early warnings



During an awareness program for the public on community based landslide earl y warning approach.

2. ***Site Specific landslide risk investigations:***

Regional offices NBRO established in landslide prone districts have been enabled with emergency dispatch teams for promptly responding to landslides and cutting failures that may occur due to rainfall. These NBRO officials will arrive and investigate the sites upon request and provide proper recommendations to minimize any risks associated. Facilitations for efficient, speedy and real-time involvement of the NBRO officers in the field have been such strengthened. Moreover, special precautionary measures have been introduced NBRO field officers with regard to the prevailing Covid-19 pandemic situation whilst carrying out the fieldworks uninterrupted. The way to request NBRO services at ground instability site would be by informing the incident to Grama Niladhari of one's area which will be followed by a subsequent request letter issued to the NBRO by the relevant Divisional Secretary or else, the NDRSC (National Disaster Relief Services Center) officer working attached to the respective Divisional Secretariat



Geologists of NBRO during field inspections

It should be mentioned that the intention of NBRO with such strong preparedness is to provide an efficient and prompt service to the public. Hence some extent of preparedness from the public side is as important. It is essential that you be mindful about ground instabilities in your surrounding and the driving causes behind them. Therefore, if you are reading this article it is important that you pay special attention to following information.

A landslide can be defined as movement of a mass of rock, soil or debris downslope under gravity. This movement could be a flow, creep, slide or a fall Landslide is a natural phenomenon resulted by unstable rock and soil layers. Soil layers generally become unstable due to high thickness or in other words height. The thickness of soil layers increase as the underlying rock weathers and produce more soil. Weathering of rocks is facilitated by optimum amount of rain and environmental temperature. That is why tropical countries tend to experience more landslides. Unstable soil layers waiting to trigger movement will receive the necessary thrust from the rains at some point. Therefore, landslides will most likely happen during monsoon season as we know. We should also keep in mind the influence of inter-monsoon rains as well as marginal

weather changes in surrounding oceans experienced by Sri Lanka owing to it being an island

It is obvious that we cannot completely escape landslide threat. However it is possible for us to reduce the landslide risk. This is where the "Preparedness" we discussed about so far, becomes useful. It is valuable if people can recognise landslide risks in their surroundings in an early stage. For this it is important that they know what symptoms to look for which have been briefly discussed below.

Landslide cracks: These can be also called ground cracks or tension cracks. When an unstable chunk of soil is about to break apart from the stable soil mass, the so called tension cracks will start to develop. It is possible to observe this symptom in each and every landslide. Hence, identifying this symptom will be like getting a warning sign. In addition, unusual cracks developing on buildings are also a pre-landslide indicator. Starting from the floor, such cracks may develop upwards along the walls of a building. As the width of such landslide cracks is higher closer to the ground and lessen as the crack develop upwards we can easily differentiate them from cracks developed due to structural defects of the buildings which usually develop in opposite (top to ground) direction on the walls.



Tension cracks developed in landslide locations

Sometimes, rainwater infiltrating into these tension cracks developed in the crown area of the landslide, may form underground waterways whilst dissolving soil particles and eventually result in muddy water springs downslope. Due to displacement of soil layers, underground waterways can sometimes be blocked resulting in drying up of existing water springs. Such blocked waterways may sometimes find a different paths to travel and emerge as new springs. We should pay special attention to unusually muddy water springs rather than clear water springs because muddy water is an indication that there is some kind of erosion happening inside the ground. Apart from aforementioned, tilting of fences, trees or telephone/electrical posts, dead trees, cracked/subsided roads, unusual bulging or subsidence of ground, unusual variations of water level of streams, displacement of concrete rings in dug wells, broken underground pipelines, offset door and window panes could all be indications of an impending landslide.

So if you are living in a landslide prone or mountainous area, it is important that you commit above-mentioned matters to your memory. Even otherwise, knowing these information could turn out to be useful to each of us at some point of our lives.

Paying attention to your surrounding environment is essential if you want to be able to



Some of pre-landslide symptoms

identify mentioned landslide symptoms. It is really important to practice preparedness strategies in a day to day basis, especially if you are living in a mountainous terrain.

Therefore,

Before rainy season:

- Clean, repair and maintain surface rain water drains around your house premises and within cultivating lands so that rainwater may flow down without any blockage.
- Seal any water retaining spots in your home garden.
- If there are any tension cracks on the ground, seal those cracks using clayey soil so that rain water would not flow into them.
- If there is an unstable earth cut near your home, you should stop rainwater flowing down the upper slope from infiltrating into the cut gushing onto it. Divert the flow away from the cut by constructing a contour drain (parallel to the cut and above the cut) allowing water to cascade down a different path away from the cut.

- Higher the unstable cut, higher the tendency to failure. If the cut is constitutes predominantly of soil, it is even more risky. Hence, take action to stabilize the cut according to engineering advice.
- Support local authorities to establish proper land use patterns in the area by adopting proper land use practices yourself.
- Make early plans on how to act in an emergency with the coordination of Local Authorities and NBRO District Offices.



Proper surface water management in cultivation lands and near earth cuts

During intense rains:

- Stay alert and awake. Many landslide fatalities occur while people are asleep.
- Be vigilant about unusual sounds that might indicate moving debris, such as trees cracking or boulders knocking together.
- Do not mock if someone senses a state of danger. Consider evacuating from the place.
- If there are unstable earth cuts near your house, refrain from staying/sleeping is rooms/parts of the house adjacent to it. Keep away as far as possible from the unstable cut.
- Be especially alert while driving through landslide susceptible mountainous areas. Watch the road carefully and do not linger close to risky places (ex: near unstable road cuts)
- After heavy rains, look for any signs of landslides on upper and lower bounds of the slopes in your area.

- Pay attention to landslide early warnings issued by NBRO in coordination with the Disaster Management Center and respond accordingly.

If you suspect immediate landslide danger:



How rooms adjacent to unstable cuts have been damaged

- Inform nearby residents promptly.
- Evacuate from the risk area to a safer location immediately.
- If possible, make a written notice about the danger and put it on display at the place.
- Inform local officials (GN) and contact a consulting agency (DMC, NBRO) or police as soon as possible.
- Be vigilant even after an intense rain has stopped. Landslides can take place several hours or may be even several days after. Therefore, even if the intense rain has stopped, refrain from going back inside houses located in the risk zone until experts come, inspect and confirm that it is safe to do so.

-National Building Research Organisation.